

Study 1b Regression Analysis

Load the data

```
data<- read.csv("/Users/yaldasmac/Desktop/cleaned_data.csv")

# Load required libraries
library(nnet)      # For multinomial logistic regression
library(effects)   # For plotting effects
```

Loading required package: carData

lattice theme set by effectsTheme()
See ?effectsTheme for details.

```
library(ggplot2)  # For visualization
library(car)      # For Anova function
```

Registered S3 method overwritten by 'car':
 method from
 na.action.merMod lme4

```
library(emmeans)  # For estimated marginal means
```

Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'

```
library(dplyr)
```

Attaching package: 'dplyr'

The following object is masked from 'package:car':

```
recode
```

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

```
intersect, setdiff, setequal, union
```

```
library(tidyr)
```

Data Preparation

```
# Create moral foundation type variable (individualizing vs. binding)
data$moral_foundation_type <- ifelse(data$care == 1 | data$fairness == 1,
                                     "Individualizing",
                                     ifelse(data$purity == 1 | data$loyalty == 1
                                             | data$authority == 1,
                                             "Binding", NA))

# Ensure Label is a factor with "Only Prolife" as the reference level
data$stance <- factor(data$Label,
                      levels = c("Only Prolife",
                                  "Prolife talking about Prochoice",
                                  "Prochoice talking about Prolife",
                                  "Only Prochoice"))

# Convert moral_foundation_type to factor
data$moral_foundation_type <- factor(data$moral_foundation_type)
```

Multinomial Logistic Regression Model with main effects only

```
# Define the model
model1 <- multinom(stance ~ moral_foundation_type + sentiment_score, data = data)
```

```
# weights:  16 (9 variable)
initial  value 235958.390617
iter   10 value 225282.813666
final   value 225082.706656
converged
```

```
# Display model summaries
summary(model1)
```

Call:

```
multinom(formula = stance ~ moral_foundation_type + sentiment_score,
          data = data)
```

Coefficients:

	(Intercept)
Prolife talking about Prochoice	-0.8464497
Prochoice talking about Prolife	-0.5029017
Only Prochoice	-1.5050929
	moral_foundation_typeIndividualizing
Prolife talking about Prochoice	0.6154509
Prochoice talking about Prolife	0.8850781
Only Prochoice	1.3418277
	sentiment_score
Prolife talking about Prochoice	-0.54652321
Prochoice talking about Prolife	-0.73622146
Only Prochoice	0.07596918

Std. Errors:

	(Intercept)
Prolife talking about Prochoice	0.03800024
Prochoice talking about Prolife	0.03413618
Only Prochoice	0.04692993
	moral_foundation_typeIndividualizing
Prolife talking about Prochoice	0.03880818
Prochoice talking about Prolife	0.03480210

Only Prochoice 0.04749825

	sentiment_score
Prolife talking about Prochoice	0.01254937
Prochoice talking about Prolife	0.01091203
Only Prochoice	0.01232374

Residual Deviance: 450165.4

AIC: 450183.4

```
# Calculate z-values and p-values (multinom doesn't provide these automatically)
z_values <- summary(model1)$coefficients / summary(model1)$standard.errors
p_values <- (1 - pnorm(abs(z_values))) * 2
print(p_values)
```

	(Intercept)
Prolife talking about Prochoice	0
Prochoice talking about Prolife	0
Only Prochoice	0

	moral_foundation_typeIndividualizing
Prolife talking about Prochoice	0
Prochoice talking about Prolife	0
Only Prochoice	0

	sentiment_score
Prolife talking about Prochoice	0.000000e+00
Prochoice talking about Prolife	0.000000e+00
Only Prochoice	7.072536e-10

Model with Interaction

```
# Interaction model
model2 <- multinom(stance ~ moral_foundation_type * sentiment_score, data = data)
```

```
# weights: 20 (12 variable)
initial value 235958.390617
iter 10 value 226301.168880
final value 225007.723305
converged
```

```
# Display model summaries
summary(model2)
```

Call:

```
multinom(formula = stance ~ moral_foundation_type * sentiment_score,
  data = data)
```

Coefficients:

	(Intercept)	
Prolife talking about Prochoice	-0.7441994	
Prochoice talking about Prolife	-0.4163190	
Only Prochoice	-1.3339700	
	moral_foundation_typeIndividualizing	
Prolife talking about Prochoice		0.513834
Prochoice talking about Prolife		0.799432
Only Prochoice		1.170072
	sentiment_score	
Prolife talking about Prochoice	-1.1020948	
Prochoice talking about Prolife	-1.3405173	
Only Prochoice	-0.6642977	
	moral_foundation_typeIndividualizing:sentiment_score	
Prolife talking about Prochoice		0.5781803
Prochoice talking about Prolife		0.6266439
Only Prochoice		0.7625772

Std. Errors:

	(Intercept)	
Prolife talking about Prochoice	0.03990867	
Prochoice talking about Prolife	0.03626715	
Only Prochoice	0.04960321	
	moral_foundation_typeIndividualizing	
Prolife talking about Prochoice		0.04065141
Prochoice talking about Prolife		0.03688181
Only Prochoice		0.05016979
	sentiment_score	
Prolife talking about Prochoice	0.06998381	
Prochoice talking about Prolife	0.06410015	
Only Prochoice	0.08582039	
	moral_foundation_typeIndividualizing:sentiment_score	
Prolife talking about Prochoice		0.07113931
Prochoice talking about Prolife		0.06505269
Only Prochoice		0.08672270

Residual Deviance: 450015.4
AIC: 450039.4

```
# Calculate z-values and p-values
z_values <- summary(model2)$coefficients / summary(model2)$standard.errors
p_values <- (1 - pnorm(abs(z_values))) * 2
print(p_values)
```

	(Intercept)	
Prolife talking about Prochoice	0	
Prochoice talking about Prolife	0	
Only Prochoice	0	
	moral_foundation_typeIndividualizing	
Prolife talking about Prochoice	0	
Prochoice talking about Prolife	0	
Only Prochoice	0	
	sentiment_score	
Prolife talking about Prochoice	0.000000e+00	
Prochoice talking about Prolife	0.000000e+00	
Only Prochoice	9.992007e-15	
	moral_foundation_typeIndividualizing:sentiment_score	
Prolife talking about Prochoice	4.440892e-	
16		
Prochoice talking about Prolife	0.000000e+00	
Only Prochoice	0.000000e+00	

Model Comparison

```
# Compare models with and without interaction
anova(model1, model2)
```

Likelihood ratio tests of Multinomial Models

Response: stance

	Model	Resid. df	Resid. Dev	Test	Df
1	moral_foundation_type + sentiment_score	510615	450165.4		
2	moral_foundation_type * sentiment_score	510612	450015.4	1 vs 2	3
	LR stat. Pr(Chi)				
1					
2	149.9667	0			

```
# Compare models
AIC(model1, model2)
```

```
      df      AIC
model1  9 450183.4
model2 12 450039.4
```

```
BIC(model1, model2)
```

```
      df      BIC
model1  9 450273.8
model2 12 450160.0
```

Visualization-Interaction Effect Between Moral Foundations and Sentiment on Stance

```
# Create a prediction grid
moral_types <- c("Binding", "Individualizing")
sentiment_points <- c(-0.75, -0.25, 0.25, 0.75) # Representative sentiment values
newdata <- expand_grid(
  moral_foundation_type = moral_types,
  sentiment_score = sentiment_points
)

# Generate predictions from your model
pred_probs <- predict(model2, newdata = newdata, type = "probs")
newdata <- cbind(newdata, as.data.frame(pred_probs))

# Reshape and prepare for plotting
plot_data <- newdata %>%
  pivot_longer(
    cols = c("Only Prolife", "Prolife talking about Prochoice",
             "Prochoice talking about Prolife", "Only Prochoice"),
    names_to = "stance",
    values_to = "probability"
  ) %>%
  mutate(sentiment_group = factor(
    case_when(
      sentiment_score == -0.75 ~ "Strong Negative (-0.75)",
```

```

    sentiment_score == -0.25 ~ "Moderate Negative (-0.25)",
    sentiment_score == 0.25 ~ "Moderate Positive (0.25)",
    sentiment_score == 0.75 ~ "Strong Positive (0.75)"
  ),
  levels = c("Strong Negative (-0.75)", "Moderate Negative (-0.25)",
             "Moderate Positive (0.25)", "Strong Positive (0.75)")
))

# Define your custom colors for each stance
custom_colors <- c(
  "Only Prolife" = "#B31529",
  "Prolife talking about Prochoice" = "#F6A482",
  "Prochoice talking about Prolife" = "#8EC4DE",
  "Only Prochoice" = "#1065AB"
)

# Create the improved visualization with manual colors
ggplot(plot_data, aes(x = factor(moral_foundation_type, levels = moral_types), y = probability)) +
  geom_bar(stat = "identity", position = "stack", color = "white", size = 0.2) +
  facet_wrap(~ sentiment_group, nrow = 1) +
  labs(
    title = "Sentiment Moderates the Effect of Moral Foundations on Abortion Stance",
    subtitle = "Predicted probability of each stance by moral foundation type across sentiment",
    x = "Moral Foundation Type",
    y = "Predicted Probability",
    fill = "Stance",
    caption = "Note: Sentiment scores range from -1 (extremely negative) to 1 (extremely positive)"
  ) +
  scale_fill_manual(values = custom_colors) +
  theme_minimal(base_size = 12) +
  theme(
    legend.position = "bottom",
    strip.text = element_text(face = "bold", size = 10),
    plot.title = element_text(size = 14, face = "bold"),
    plot.subtitle = element_text(size = 11),
    panel.grid.major.x = element_blank(),
    panel.spacing = unit(1.5, "lines"),
    axis.text.x = element_text(angle = 0, size = 10),
    plot.caption = element_text(hjust = 0, size = 9, face = "italic")
  )

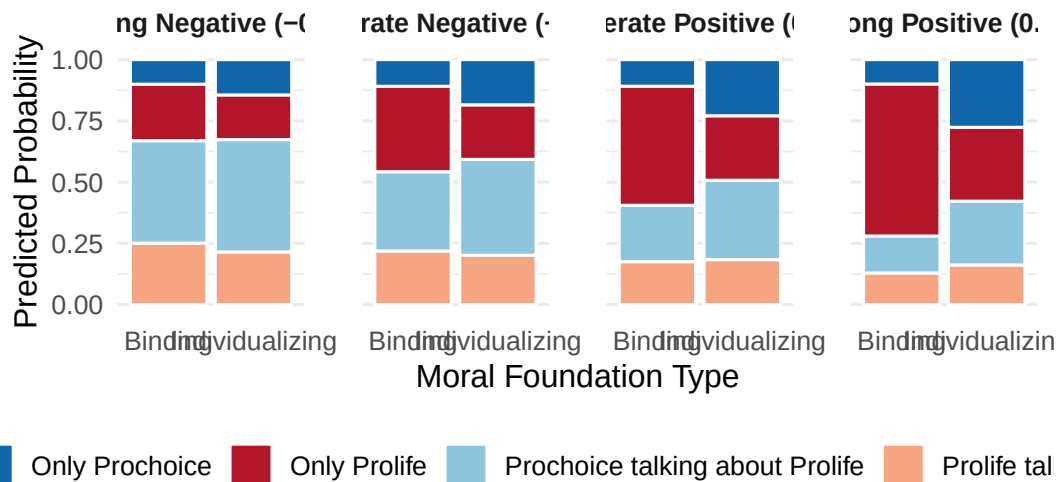
```

Warning in geom_bar(stat = "identity", position = "stack", color = "white", :

Ignoring unknown parameters: `size`

Sentiment Moderates the Effect of Moral Foundations

Predicted probability of each stance by moral foundation type across sentiment



Note: Sentiment scores range from -1 (extremely negative) to 1 (extremely positive).

```
# Save the plot
ggsave("stance_by_moral_sentiment.jpg",
       width = 10, height = 5, dpi = 300)
```

Visualization-Main effect of Morality

```
eff <- Effect("moral_foundation_type", model1)

me <- data.frame(
  foundation = rep(as.character(eff$x$moral_foundation_type), times = ncol(eff$prob)),
  stance     = rep(colnames(eff$prob), each = nrow(eff$prob)),
  predicted  = as.vector(eff$prob),
  conf.low   = as.vector(eff$lower.prob),
  conf.high  = as.vector(eff$upper.prob)
)

# effects returns "prob.Only.Prolife" etc. - strip "prob." and turn dots back to spaces
me$stance <- gsub("\\.", " ", sub("^prob\\.", "", me$stance))
```

```

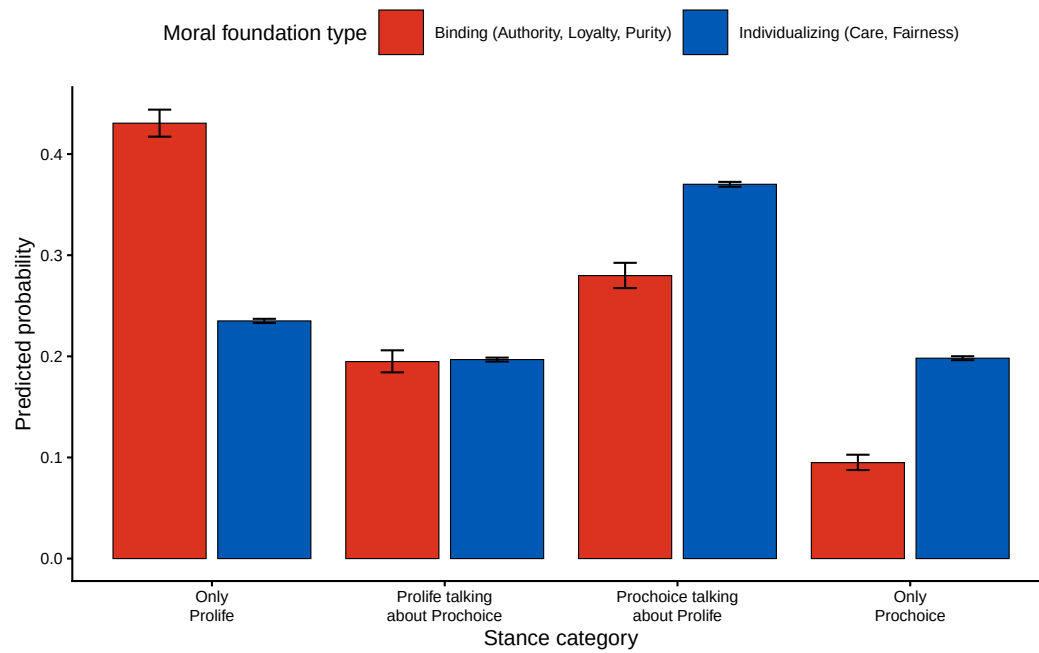
pref <- c("Only Prolife", "Prolife talking about Prochoice",
          "Prochoice talking about Prolife", "Only Prochoice")
stopifnot(all(pref %in% me$stance)) # fail loudly rather than produce silent NAs

me$stance <- factor(
  me$stance, levels = pref,
  labels = c("Only\nProlife", "Prolife talking\nabout Prochoice",
             "Prochoice talking\nabout Prolife", "Only\nProchoice")
)
me$foundation <- factor(me$foundation, levels = c("Binding", "Individualizing"))

p <- ggplot(me, aes(x = stance, y = predicted, fill = foundation)) +
  geom_col(position = position_dodge(width = 0.9),
           width = 0.8, color = "black", linewidth = 0.2) +
  geom_errorbar(aes(ymin = conf.low, ymax = conf.high),
               width = 0.2, position = position_dodge(width = 0.9)) +
  scale_fill_manual(
    values = c("Binding" = "#DC3220", "Individualizing" = "#005AB5"),
    labels = c("Binding (Authority, Loyalty, Purity)",
               "Individualizing (Care, Fairness)")
  ) +
  labs(x = "Stance category", y = "Predicted probability",
       fill = "Moral foundation type") +
  theme_classic(base_size = 8) +
  theme(legend.position = "top", panel.grid.major.x = element_blank())

print(p)

```



```
ggsave("morality_main_effect.pdf", plot = p, width = 7.2, height = 3.5)
```